Rotational motion CHEAT SHEET

CENTRE OF MASS

$X = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$ (a weighted average)

2 kg at 0, 6 kg at 0.8 m → X = 4.8/8 = 0.6 m

equal masses → the middle; 600 kg → 0.797 m

many: X = Σmx/M, Y = Σmy/M; a uniform rod: its middle

M A = F_{ext}: internal forces cancel in pairs

firework burst at the top: halves land at 39.8 and 119.3 m, centre of mass at 79.5 m, the unbroken shell's range

P = M V; no external force → P fixed, X stays put

SIMULATIONS

every simulation checked against its formula:
firework (centre of mass) · puck (L fixed) ·
fan (300 rad, 47.75 turns) · bucket (a = 3.27 m/s²)

CROSS PRODUCT AND ANGULAR VELOCITY

|A × B| = AB sin θ, at right angles to both (right hand)

4 and 3 at 30° → 6; parallel → 0; B × A = −A × B

i × j = k, j × k = i, k × i = j; i × i = 0

$(P_x i + P_y j) \times (Q_x i + Q_y j) = (P_x Q_y - P_y Q_x)\,k$

$(2i + j) \times (3i + 4j) = 5k$

θ in radians = arc/r; one turn = 2π rad

ω = Δθ/Δt; v = ωr; α = Δω/Δt

fan 300 rpm = 31.42 rad/s; tip at 0.6 m: 18.85 m/s

TORQUE AND ANGULAR MOMENTUM

τ = r × F, τ = rF sin θ = F × lever arm

door: 10 N at 0.9 m → 9 N m; at 0.1 m → 1 N m; 30° → 4.5

in components: $\tau = xF_y - yF_x$ (anticlockwise +)

l = r × p = p × lever arm; free particle: l fixed

τ = dl/dt; system: $\tau_{ext} = dL/dt$

no outside torque → L can't change

puck: 2 m/s at 0.5 m → 4 m/s at 0.25 m (mvr = 0.2)

EQUILIBRIUM

equilibrium: ΣF = 0 and Στ = 0

couple: F × d, the same about any point (15 × 0.36 = 5.4)

moments: $m_1 g\, d_1 = m_2 g\, d_2$; 30 kg at 1.6 m ↔ 60 kg at 0.8 m

lever: 900 N at 0.1 m ← 100 N at 0.9 m; MA = 9

plank 4 m, 20 kg, 60 kg at 1 m: R_L 539 N, R_R 245 N

centre of gravity: total gravity torque zero = X in uniform g

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MOMENT OF INERTIA

$K = \frac{1}{2}I\omega^2$, $I = \Sigma mr^2$ (mass counts by r^2)

2×2 kg at 0.3 m: $I = 0.36$, 18 J at 10 rad/s; at 0.6 m: $\times 4$

ring MR^2 · disc, solid cylinder $\frac{1}{2}MR^2$

rod: middle $\frac{ML^2}{12}$, end $\frac{ML^2}{3}$

solid ball $\frac{2}{5}MR^2$ · thin hollow ball $\frac{2}{3}MR^2$

$k = \sqrt{(I/M)}$: disc $R/\sqrt{2} \approx 0.71R$; rod (middle) $L/\sqrt{12}$

slices: $N = 2, 10, 100 \rightarrow 0.0625, 0.0825, 0.083325$ ($\rightarrow 1/12$)

CHECK BEFORE YOU TRUST IT

before you trust an answer:

1 angles in radians? speeds in rad/s, not rpm?

2 about which axis? (I and τ both depend on it)

3 is there an outside torque? if not, L is kept

4 find the straight-line twin, and swap the letters

THE TURNING LAWS

$\omega = \omega_0 + \alpha t$; $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$; $\omega^2 = \omega_0^2 + 2\alpha\theta$

fan 30 rad/s, $\alpha = -1.5$: stops at 20 s after 300 rad

$W = \tau\theta$; $P = \tau\omega$; $\tau = I\alpha$

engine 200 N m at 3,000 rpm → 62.83 kW

bucket on a drum: $a = \frac{mg}{m + I/R^2} = 3.27$ m/s²

$L = I\omega$; no outside torque → $I_1\omega_1 = I_2\omega_2$

chair: I 4 → 1.6, ω 1.5 → 3.75 rad/s (energy 4.5 → 11.25 J)

STRAIGHT LINE AND TURNING

straight line	turning
$x \cdot v \cdot a$	$\theta \cdot \omega \cdot \alpha$
$m \cdot F$	$I \cdot \tau$
$F = ma$	$\tau = I\alpha$
$\frac{1}{2}mv^2 \cdot Fs \cdot Fv$	$\frac{1}{2}I\omega^2 \cdot \tau\theta \cdot \tau\omega$
$p = mv$, $F = dp/dt$	$L = I\omega$, $\tau = dL/dt$
p kept if $\Sigma F = 0$	L kept if $\Sigma\tau = 0$

A FLYWHEEL FOR THE TRAM

tram 30 t at 12 m/s: $\frac{1}{2}mv^2 = 2.16$ MJ; ¾ of it: 1.62 MJ

a wheel of 500 kg, $R = 0.6$ m, its rim under 100 m/s:

$\frac{1}{2}I\omega^2 = E \rightarrow \omega = \sqrt{(2E/I)}$; rim speed $v = \omega R$

disc $\frac{1}{2}MR^2 = 90$ kg m²: 189.74 rad/s, rim 113.84 m/s: too fast

ring $MR^2 = 180$ kg m²: 134.16 rad/s (1,281 rpm), rim 80.50 m/s

the ring wins: mass far out holds the energy at less spin